BIO 005 Human Physiology · Week 8 notes · Dr. Sharilyn Rennie

# The reproductive systems

The need-to-know anatomy of both systems, the gonadal axis and its feedback, testis structure, spermatogenesis and testosterone, the path of sperm, semen and the erection and ejaculation reflexes, oogenesis and follicle growth, the ovarian and uterine cycles with the switch to positive feedback, the actions of estrogen and progesterone, fertilization, implantation and the hormones of pregnancy, labor, lactation, menopause and sexually transmitted infections. Week 8 competencies 16 to 22.

**To print:** these notes as a two-column PDF.

**How to use these notes.** Read these notes alongside your textbook reading for the week, then use them to review. Each part names where to read in Silverthorn and in OpenStax, and links Dr. Rennie's videos where there is one.

**Your textbook:** Silverthorn Chapter 26, Reproduction and Development. **Using OpenStax instead?** Anatomy and Physiology 2e, Chapter 27, The Reproductive System, and Chapter 28, Development and Inheritance.

## Need-to-know anatomy

**In your textbook:** Silverthorn Chapter 26, the parts on male reproduction and female reproduction, on the anatomy of each system.

Each system has gonads that make gametes and hormones, ducts, accessory glands and external genitalia. Many male and female structures grow from the same embryonic tissue (they are homologous).

### Male structures

**Scrotum**

Skin pouch that holds the testes outside the pelvic cavity, where they stay cooler. Its dartos muscle (smooth muscle) wrinkles the skin; the cremaster muscle (skeletal muscle in the spermatic cord) pulls the testes up.

**Testis**

A fibrous capsule (tunica albuginea) divides it into 200 to 300 lobules, each with one to three coiled seminiferous tubules where sperm are made. Leydig cells sit between the tubules. The tubules drain into the rete testis.

**Epididymis**

A coiled duct along the back of each testis where sperm mature and are stored.

**Ductus (vas) deferens**

Muscular duct from the epididymis, through the spermatic cord and inguinal canal, behind the bladder. Its widened end is the ampulla.

**Spermatic cord**

Carries the ductus deferens, testicular artery, pampiniform plexus of veins, nerves, lymphatics and the cremaster.

**Ejaculatory duct**

Ampulla plus the seminal vesicle duct. It runs through the prostate into the urethra.

**Urethra**

Shared by urine and semen: prostatic, intermediate (membranous) and spongy parts, ending at the external urethral orifice.

**Accessory glands**

Paired seminal vesicles behind the bladder, the prostate around the urethra below the bladder, and pea-sized bulbourethral (Cowper's) glands below the prostate.

**Penis**

Two corpora cavernosa on top and one corpus spongiosum below, which carries the urethra and ends as the glans. Erectile tissue is blood spaces (sinuses) surrounded by smooth muscle. The

bulbospongiosus and ischiocavernosus, skeletal muscles at its root, push semen out.

### Female structures

#### Ovary

Almond-sized gonad beside the uterus, held by the ovarian ligament (to the uterus), the suspensory ligament (to the pelvic wall, carrying the ovarian vessels) and a fold of the broad ligament. The cortex holds the follicles; the medulla holds vessels and nerves.

#### Uterine (fallopian) tube

Its funnel (infundibulum) opens into the pelvic cavity, fringed by fimbriae that sweep over the ovary. Fertilization usually happens in the ampulla, the widest part; the narrow isthmus joins the uterus. Ciliated cells and smooth muscle move the oocyte toward the uterus, and peg cells secrete nourishing fluid.

#### Uterus

Muscular organ between the bladder and rectum, normally tipped forward over the bladder (anteflexed) and held by the broad, round, uterosacral and cardinal ligaments. Parts: fundus, body and cervix. The cervical canal opens at the internal os (to the uterus) and the external os (to the vagina). Cervical glands make mucus that changes with the cycle.

#### Wall of the uterus

Perimetrium (outer serous layer), myometrium (smooth muscle that contracts in labor), and endometrium (lining). The endometrium's basal layer is permanent; its functional layer thickens each cycle and is shed. Spiral arterioles supply the functional layer and change with the hormones.

#### Vagina

Muscular canal from cervix to outside, with no glands. Lactobacilli ferment glycogen from shed cells into lactic acid, so it is acidic: protective against infection but hard on sperm.

#### Vulva

External genitalia: mons pubis, labia, clitoris (erectile tissue), vestibule with the urethral and vaginal openings, bulbs of the vestibule (erectile tissue), and greater vestibular (Bartholin's) glands that add mucus during arousal.

#### Mammary gland

A modified sweat gland. Lobes divide into lobules of milk-making alveoli wrapped by myoepithelial cells; lactiferous ducts carry milk to the nipple.

| Male structure | Female homolog |
|---|---|
| **Testis** | Ovary |
| **Penis (corpora cavernosa and glans)** | Clitoris |
| **Corpus spongiosum and bulb of the penis** | Bulbs of the vestibule |
| **Scrotum** | Labia majora |
| **Prostate** | Paraurethral (Skene's) glands |
| **Bulbourethral glands** | Greater vestibular glands |

## The gonadal axis

**Competency 16:** Hypothalamic pituitary gonadal axis.

**In your textbook:** Silverthorn Chapter 26, the part on basic patterns of reproduction, on hormonal control of reproduction.

Reproduction uses the same three-level pattern as the thyroid and adrenal axes: hypothalamus, anterior pituitary, then a gland whose hormones feed back. The same hormones run both sexes; the gonads respond differently.

### The gonadal axis, step by step

1. Neurons in the hypothalamus release gonadotropin-releasing hormone (GnRH) in pulses into the hypophyseal portal vessels.
2. GnRH reaches the gonadotrophs of the anterior pituitary, which release luteinizing hormone (LH) and follicle-stimulating hormone (FSH) into the blood.
3. LH and FSH travel to the gonads, the testes or the ovaries.
4. LH acts on the cells that make steroid hormones: Leydig cells in the testis, theca cells in the ovary.
5. FSH acts on the cells that support the gamete: Sertoli cells in the testis, granulosa cells in the ovary.
6. The gonad releases its steroid hormones (testosterone, or estrogen and progesterone) and the protein hormone inhibin into the blood.
7. The steroids feed back on both the hypothalamus (less GnRH) and the anterior pituitary (less LH and FSH). Inhibin feeds back on the anterior pituitary and lowers FSH alone.

8. This negative feedback keeps levels steady, except at midcycle in women, when high, sustained estrogen switches to positive feedback and sets off the LH surge.

| Hormone | Made by | Feeds back on | Effect |
|---|---|---|---|
| **Testosterone** | Leydig cells | Hypothalamus and anterior pituitary | Less GnRH and less LH |
| **Inhibin** | Sertoli cells; granulosa cells and the corpus luteum | Anterior pituitary | Less FSH |
| **Estrogen, low to moderate** | Growing follicles | Hypothalamus and anterior pituitary | Less GnRH, LH and FSH |
| **Estrogen, high for about two days** | Dominant follicle | Hypothalamus and anterior pituitary | More GnRH; the LH surge (positive feedback) |
| **Progesterone with estrogen** | Corpus luteum | Hypothalamus and anterior pituitary | Strong suppression of GnRH, LH and FSH |

**Gonadotropins**

LH and FSH, the two anterior pituitary hormones that act on the gonads. Both are glycoproteins made by the same cells, the gonadotrophs.

**Pulsatile GnRH**

GnRH comes in bursts, and the pituitary needs the bursts. Steady GnRH down-regulates the GnRH receptors, and after a brief rise LH and FSH fall. Long-acting drugs that mimic GnRH use this to shut the gonads down, for example to lower testosterone in some prostate cancers.

**Inhibin**

A protein hormone from the FSH-driven cells (Sertoli and granulosa cells) that selectively lowers FSH. This lets FSH and LH be adjusted separately even though one releasing hormone drives both.

### Puberty

- In childhood the hypothalamus is very sensitive to the small amounts of sex steroids in the blood, so GnRH stays low. This is a simplified account: the brain also holds GnRH back in childhood even without the gonads.
- Near puberty the hypothalamus becomes less sensitive to that feedback, and GnRH, LH and FSH rise. In girls the adult cycle takes a few years (often about four) to settle in.
- Menarche is the first menstrual period. Early cycles often skip ovulation, and cycles usually become regular within a few years.
- Leptin from fat tissue tells the hypothalamus whether energy stores can support reproduction. It is permissive: enough leptin allows puberty and cycles, too little delays them, but it does not trigger puberty by itself.

## The testis, spermatogenesis and testosterone

**Competency 17:** Male reproductive physiology.

**In your textbook:** Silverthorn Chapter 26, the part on male reproduction, on spermatogenesis, its hormonal control and the actions of androgens.

### Keeping the testes cool

Spermatogenesis works best at about 35 °C (95 °F), a few degrees below the core temperature of 37 °C (98.6 °F). The scrotum holds the testes outside the pelvic cavity, and two muscles adjust their temperature.

- In the cold, the cremaster contracts and pulls the testes up toward the body to take up its heat. The dartos contracts and tightens the scrotal skin, which reduces heat loss.
- In the warmth, both muscles relax. The testes hang farther from the body and the loose skin loses more heat.
- In the spermatic cord, the pampiniform plexus of veins wraps the testicular artery. Cooler venous blood draws heat from the arterial blood on its way in (countercurrent heat exchange).

### Cells of the seminiferous tubule

**Spermatogenic cells**

The cells that become sperm. The youngest, the spermatogonia, sit at the outer wall of the tubule. Older cells move inward as they mature, and sperm are released into the lumen.

**Sertoli (sustentacular) cells**

Tall cells from the tubule wall to the lumen that wrap the developing sperm. They nourish them, engulf the cytoplasm spermatids shed, control their movement and release, make the fluid that carries sperm out, and

make androgen-binding protein and inhibin. FSH and testosterone act on them to drive spermatogenesis.

**Leydig (interstitial) cells**

Clusters of cells between the tubules, beside the blood vessels, that make testosterone under LH. Testosterone diffuses into the nearby tubules (a paracrine signal) and also enters the blood.

**Blood–testis barrier**

Tight junctions between neighboring Sertoli cells. Spermatogonia sit outside it, open to the blood; later germ cells develop inside it. Developing sperm carry proteins that first appear at puberty, after the immune system learned what is self, so the barrier keeps immune cells and antibodies away from them. Substances must pass through a Sertoli cell to reach them.

**Androgen-binding protein (ABP)**

A protein that Sertoli cells secrete into the tubule under FSH and testosterone. It binds testosterone and keeps its concentration very high around the developing sperm.

### Spermatogenesis, step by step

1. Primordial germ cells arrive in the testis early in embryonic life and become spermatogonia, diploid stem cells that stay dormant until puberty.
2. From puberty on, spermatogonia at the tubule wall divide by mitosis. One daughter stays a stem cell, so the supply never runs out; the other goes on to mature.
3. The maturing cell copies its DNA and grows into a primary spermatocyte, still diploid (46 chromosomes) and the largest germ cell in the tubule. Sertoli cell tight junctions open ahead of it and reseal behind it, so it crosses the blood-testis barrier without breaking it.
4. The primary spermatocyte completes meiosis I and forms two haploid secondary spermatocytes, each with 23 chromosomes. Each chromosome is still made of two copies (sister chromatids).
5. Each secondary spermatocyte completes meiosis II, which separates the copies, giving four haploid spermatids in all, each with 23 single chromosomes.
6. In spermiogenesis, each spermatid remodels into a sperm: it forms an acrosome and a tail, packs mitochondria into the midpiece, and sheds most of its cytoplasm, which a Sertoli cell engulfs.
7. The sperm is released into the lumen. It cannot swim yet. Fluid from the Sertoli cells carries it toward the epididymis.
8. The whole sequence takes a little over two months (about 64 to 74 days). The testes make hundreds of millions of sperm each day.

**Head**

Holds the nucleus with 23 tightly packed chromosomes. The acrosome, a cap-like vesicle over the front two-thirds of the nucleus, holds enzymes that digest a path to the oocyte.

**Neck and midpiece**

The neck holds centrioles that build the tail's microtubules. The midpiece holds spiral mitochondria that make ATP for swimming.

**Tail**

The flagellum, with a long principal piece and a short end piece, which whips to drive the sperm forward.

### Feedback in the male, step by step

1. LH from the anterior pituitary makes the Leydig cells secrete testosterone.
2. FSH and testosterone act on the Sertoli cells, which support spermatogenesis and secrete ABP; testosterone drives its final steps.
3. High blood testosterone inhibits GnRH from the hypothalamus and LH from the anterior pituitary, so testosterone falls back toward normal.
4. Low testosterone eases the inhibition, and GnRH, LH and testosterone rise again.
5. When sperm production is high, Sertoli cells release more inhibin, which lowers FSH.

### Testosterone from outside the body

1. High blood testosterone from injections or anabolic steroids acts as negative feedback.
2. GnRH, LH and FSH fall.
3. Without LH, the Leydig cells stop making testosterone, so the very high level inside the testis falls, even though the blood level is high.
4. Without FSH and local testosterone, the Sertoli cells cannot support spermatogenesis. Sperm production falls and the testes can shrink.

### What testosterone does

Testosterone is a steroid: it enters cells and binds androgen receptors that turn genes on and off. Some

tissues first convert it to another hormone, which is why its effects are so varied.

| Form | Made where | Main actions |
|---|---|---|
| **Testosterone** | Leydig cells (the adrenal cortex adds small amounts of weaker androgens) | Before birth, builds the male ducts and helps the testes descend. Later, drives spermatogenesis, muscle mass, sex drive (libido) and red blood cell production. |
| **Dihydrotestosterone (DHT)** | Converted from testosterone by 5-alpha reductase in target tissues | Before birth, builds the external genitalia and prostate. Later, drives prostate growth, facial and body hair, oil gland secretion (acne) and male-pattern hair loss. |
| **Estradiol** | Converted from testosterone by aromatase, in the brain, bone and fat | Closes the growth plates at the end of puberty; acts in parts of the developing brain. |

- At puberty, testosterone and DHT enlarge the penis, scrotum and accessory glands and bring on the secondary sex characteristics: muscle and bone growth, broader shoulders, facial, chest, underarm and pubic hair, thicker skin, more oil gland secretion, and a larger larynx with a deeper voice.
- Testosterone is anabolic: it increases protein synthesis, so muscle and bone mass rise. It also raises the basal metabolic rate.
- It stimulates red blood cell production (through erythropoietin), one reason men tend to have a higher hematocrit.
- Androgens support sex drive in both sexes. In women, the ovaries and adrenal cortex make small amounts.

### Clinical notes

- Cryptorchidism: the testes form near the kidneys and normally descend through the inguinal canals into the scrotum, starting in about the seventh month of pregnancy. In cryptorchidism a testis does not descend (about 3% of full-term and 30% of premature infants). Most descend on their own in the first year; if not, surgery is done, usually before 18 months of age. If both stay up, body heat destroys the developing sperm cells and causes sterility. An undescended testis also carries a several times higher risk of testicular cancer; early surgery lowers that risk but does not remove it. Testosterone is usually normal.
- Hydrocele: serous fluid in the tunica vaginalis around the testis, sometimes after injury or epididymis inflammation. It often needs no treatment.
- Varicocele: swollen testicular veins that feel like a bag of worms, more obvious when standing. Often harmless, but the pooled warm blood heats the testis, and it is a common correctable cause of a low sperm count.

## The path of sperm, semen, erection and ejaculation

**In your textbook:** Silverthorn Chapter 26, the part on male reproduction, on the accessory glands, and the part on procreation, on the male sex act.

### The path of sperm, step by step

1. Fluid from the Sertoli cells pushes sperm along the lumen of the seminiferous tubule.
2. Sperm pass through short straight tubules into the rete testis.
3. Efferent ducts carry them into the head of the epididymis, where they join the single coiled ductus epididymis.
4. Over about two weeks in the epididymis, sperm gain the ability to swim and fertilize. They are stored there and in the ductus deferens for up to several months; old sperm are reabsorbed.
5. During arousal, peristalsis in the ductus deferens moves sperm through the spermatic cord and behind the bladder to the ampulla.
6. The ampulla joins the duct of the seminal vesicle to form the ejaculatory duct, which runs through the prostate.
7. The ejaculatory duct empties into the prostatic urethra, where prostatic fluid joins.
8. Semen passes through the intermediate (membranous) urethra and into the spongy urethra, where the bulbourethral gland ducts open. (Most of their mucus was already released during arousal.)
9. Semen leaves through the external urethral orifice.

| Gland | Share of semen | What it adds | Why it matters |
|---|---|---|---|
| **Seminal vesicles** | About 60% | Alkaline, sticky fluid with fructose, prostaglandins and clotting proteins | Fructose fuels ATP production in sperm. Alkalinity neutralizes acid in the male urethra and the vagina. Prostaglandins aid sperm motility and may make the female tract's smooth muscle contract. Clotting proteins make semen gel after ejaculation. |
| **Prostate** | About 25 to 30% | Milky, slightly acidic fluid (pH about 6.5) with citrate, prostate-specific antigen (PSA) and other enzymes, and seminalplasmin | Citrate is a fuel for the Krebs cycle in sperm. PSA and other enzymes break down the clot so semen liquefies. Seminalplasmin is an antibacterial protein that kills some bacteria. |
| **Bulbourethral glands** | Small | Clear alkaline mucus, released during arousal before ejaculation | Neutralizes acid left by urine in the urethra and lubricates the end of the penis, so fewer sperm are damaged. |
| **Testes, epididymis and ductus deferens** | About 10% | Sperm and the fluid they travel in | The sperm themselves are only a few percent of the volume. |

**Semen**

Sperm plus seminal fluid from the seminiferous tubules, seminal vesicles, prostate and bulbourethral glands. Sperm are only a small share of the volume.

**Normal semen**

Typically about 2 to 5 mL per ejaculation, 50 to 150 million sperm per mL, pH 7.2 to 7.7 (slightly alkaline). The World Health Organization's lower reference limits (2021) are 1.4 mL and 16 million sperm per mL. A count below that lowers the chance of fertility, because only a tiny fraction of sperm reach the oocyte.

**Coagulation and liquefaction**

Semen gels within about 5 minutes from the seminal vesicle clotting proteins, then PSA and other prostate enzymes liquefy it after 10 to 15 minutes. If it fails to liquefy or liquefies late, sperm stay partly or fully trapped and move poorly through the cervix. The purpose of the clot is not certain.

**Capacitation**

Changes caused by secretions of the uterus and tubes: the tail beats harder, and the membrane becomes able to release the acrosome's enzymes and fuse with the oocyte. Fresh sperm cannot fertilize until this happens.

- A vasectomy cuts the ductus deferens. Semen volume barely changes, because nearly all of it comes from the glands, but it no longer carries sperm.
- Blood PSA can rise with prostate cancer but also with benign enlargement and inflammation, so a high value needs follow-up, not a conclusion.
- The prostate grows quickly from puberty to about age 30, is stable until about 45, then often enlarges and can squeeze the urethra and slow urine flow.
- Hemospermia, blood in the semen, often comes from inflammation of the seminal vesicles and is usually harmless, but it is checked to rule out infection or other causes.

## Erection, step by step

1. Touch of the penis sends sensory signals through the pudendal nerve to the sacral spinal cord. Thoughts, sights or sounds can send signals from the brain to the same center.
2. The sacral cord integrates the signals and activates parasympathetic neurons to the penis. Sympathetic signals that keep the arterioles narrow at rest are turned down.
3. Parasympathetic nerve endings and the endothelium release nitric oxide (NO).
4. NO raises cGMP in the smooth muscle of the arterioles and the erectile tissue, and the muscle relaxes.
5. The arterioles widen, and blood rushes into the sinuses of the corpora cavernosa and corpus spongiosum.
6. The swelling erectile tissue presses the veins against the tough tunica albuginea, so blood flows in faster than it can leave. The penis lengthens and stiffens.
7. The corpus spongiosum stays softer than the corpora cavernosa, so the urethra stays open for semen to pass.

### Ejaculation, step by step

1. Continued stimulation raises sensory input to reflex centers in the lower thoracic and upper lumbar cord.
2. Emission: sympathetic neurons make the smooth muscle of the epididymis, ductus deferens, ejaculatory ducts, seminal vesicles and prostate contract. Sperm and gland fluid move into the urethra.
3. Sympathetic neurons also contract the internal urethral sphincter at the base of the bladder, so semen cannot flow back into the bladder and urine does not enter the urethra.
4. Expulsion: semen filling the urethra triggers rhythmic contractions of the bulbospongiosus and ischiocavernosus, skeletal muscles driven by somatic motor neurons in the pudendal nerve. Semen is pushed out.
5. Afterward, sympathetic signals narrow the arterioles, blood drains, and the erection fades. A refractory period follows.

| Event | Division | Main effect |
|---|---|---|
| **Erection** | Parasympathetic (sacral cord), with less sympathetic tone | NO widens arterioles and erectile tissue fills |
| **Emission** | Sympathetic (thoracolumbar cord) | Smooth muscle of the ducts and glands moves semen into the urethra |
| **Closing the bladder neck** | Sympathetic | Internal urethral sphincter contracts |
| **Expulsion** | Somatic motor (pudendal nerve) | Skeletal muscle pushes semen out |

- A common memory aid: parasympathetic points, sympathetic shoots.
- Phosphodiesterase type 5 (PDE5) breaks down cGMP and ends the erection. Erectile dysfunction drugs such as sildenafil block PDE5, so cGMP lasts longer, but they need arousal to work.
- Priapism is a long, painful erection unrelated to sexual desire. It is an emergency, because trapped blood loses its oxygen and can damage the erectile tissue.
- A small amount of clear fluid can appear before ejaculation. It comes mainly from the bulbourethral glands and can carry some sperm. This is not emission, which is the first phase of ejaculation.

## Oogenesis and follicle development

**In your textbook:** Silverthorn Chapter 26, the part on female reproduction, on the ovary and oogenesis.

Oogenesis follows the same plan as spermatogenesis, mitosis and then meiosis, but on a very different schedule. All the primary oocytes a woman will ever have are made before birth, and each one pauses in meiosis for years or decades.

### Oogenesis, step by step

1. Early in fetal life, primordial germ cells migrate from the yolk sac into the developing ovaries. (The surface germinal epithelium of the ovary does not make eggs.)
2. There they become oogonia, diploid stem cells that divide by mitosis into millions of cells, about 7 million by mid-pregnancy.
3. Oogonia copy their DNA and enlarge into primary oocytes, which are still diploid. They begin meiosis I and then stop (arrest) in prophase I. No new oogonia are made after this, so there is no stem cell supply in adult life.
4. Each primary oocyte is wrapped in one layer of flat cells, forming a primordial follicle in the ovarian cortex.
5. Most die by programmed cell death (atresia). About 1 to 2 million remain at birth, roughly 300,000 to 400,000 at puberty, and only about 400 are ever ovulated.
6. After puberty, each cycle a group of follicles grows under FSH, and usually one becomes dominant.
7. Just before ovulation, the LH surge makes the primary oocyte in the dominant follicle finish meiosis I. It divides unequally into a large secondary oocyte, which keeps almost all the cytoplasm, and a tiny first polar body. Both are haploid, with 23 chromosomes that each still have two sister chromatids.
8. The secondary oocyte starts meiosis II and stops again at metaphase II. This is the cell that is ovulated.
9. If a sperm enters it, the secondary oocyte finishes meiosis II and forms the ovum and a second polar body. The nuclei of the ovum and sperm then join to make a diploid zygote.
10. If no sperm arrives, the secondary oocyte degenerates within about a day. The first polar body may also divide, and all polar bodies degenerate, so one primary oocyte yields one egg.

- The unequal divisions give the egg nearly all the cytoplasm and nutrients, which the embryo lives on until it reaches the uterus and implants.

- An ovulated oocyte has waited in prophase I since before birth, 12 to 50 years, one reason chromosome errors in eggs rise with the mother's age.

| | Spermatogenesis | Oogenesis |
|---|---|---|
| **When it happens** | Puberty to old age | Starts before birth; ovulation from puberty to menopause |
| **Time to make one gamete** | About 64 to 74 days | 12 to 50 years, from fetal life to ovulation |
| **Pauses in meiosis** | None | Prophase I for years; metaphase II until fertilization |
| **Gametes from one meiosis** | Four equal sperm | One large egg and two or three polar bodies |
| **Gametes in a lifetime** | More than a trillion | About 400 to 500 |
| **Support cells** | One Sertoli cell supports many developing sperm | Many granulosa cells support one oocyte |
| **Chromosome errors** | Less common | More common, and rising with the mother's age |

### Follicle development, step by step

1. A primordial follicle is recruited: its flat cells become cuboidal and the oocyte enlarges. It is now a primary follicle.
2. The oocyte secretes a thick, clear glycoprotein coat, the zona pellucida, between itself and the surrounding cells.
3. The surrounding cells multiply into several layers, now called granulosa cells. With more than one layer, it is a secondary follicle.
4. Granulosa cells connect to the oocyte by gap junctions that pass ions, metabolites and signals, and the two also exchange paracrines. These local signals make the oocyte grow and coordinate their development.
5. Connective tissue cells around the follicle form the theca. The theca interna is rich in blood vessels and makes androgens under LH; the theca externa is an outer fibrous layer.
6. Fluid collects between the granulosa cells and merges into one cavity, the antrum. From here on (the antral or vesicular follicle), further growth needs FSH and LH.
7. The antrum expands until the oocyte, in its ring of granulosa cells (the corona radiata), sits on a stalk to one side. The mature (Graafian) follicle bulges from the ovary surface.
8. The follicle ruptures and releases the secondary oocyte with its corona radiata. The remaining follicle becomes the corpus luteum.

**Preantral phase**

Primordial, primary and secondary follicles. Growth is controlled mainly by paracrine signals inside the ovary and does not depend on FSH and LH.

**Antral phase**

Follicles with an antrum. They depend on FSH and LH, grow quickly, and compete; one becomes dominant.

**Atresia**

Programmed death of a follicle. It can happen at any stage, and more than 99% of follicles end this way.

**Corpus luteum**

The yellow gland formed from the ruptured follicle. It makes progesterone, estrogen, inhibin and relaxin. When it breaks down it leaves a white scar, the corpus albicans.

- Growing a follicle from recruitment to ovulation takes months, close to a year by some estimates.
- Older ovaries look scarred and pitted from years of ovulation.

## The ovarian and uterine cycles

**Competencies 18, 19 and 22:** Ovarian cycle; Uterine cycle; Hormone cycle graph interpretation.

**In your textbook:** Silverthorn Chapter 26, the part on female reproduction, on the menstrual cycle and its hormonal control.

The ovarian cycle grows a follicle, releases an oocyte and forms a corpus luteum. The uterine cycle builds and sheds the endometrium, driven by the ovary's hormones, so the two run in step. A textbook cycle is 28 days, but adult cycles normally run about 21 to 35 days.

### The ovarian cycle and its hormones, step by step

1. At the end of the previous cycle the corpus luteum dies, estrogen, progesterone and inhibin fall, and with less negative feedback FSH rises a little.
2. Day 1 is the first day of bleeding. The rising FSH rescues a group of antral follicles from atresia, and they begin to grow.

3. LH drives the theca cells to make androgens. FSH drives the granulosa cells to convert those androgens to estradiol with the enzyme aromatase. Very little of the androgen reaches the blood.
4. Estradiol rises. At low to moderate levels, with inhibin from the granulosa cells, it is negative feedback on the hypothalamus and anterior pituitary, so FSH falls in mid follicular phase.
5. Only the follicle most sensitive to FSH (most FSH receptors, best blood supply) survives the fall and becomes dominant; the others undergo atresia. Its granulosa cells gain LH receptors.
6. The dominant follicle makes more and more estradiol. When estradiol stays high for about two days, its feedback switches from negative to positive.
7. The hypothalamus releases more GnRH, the pituitary becomes more sensitive to it, and the anterior pituitary releases a large LH surge with a smaller rise in FSH, around day 13.
8. The LH surge makes the primary oocyte finish meiosis I and starts an inflammation-like reaction in the follicle wall: vessels leak, enzymes (metalloproteinases) weaken the wall, and blood flow stops at the bulging spot, which thins.
9. Ovulation: roughly a day and a half (about 32 to 36 hours) after the surge begins, around day 14, the follicle ruptures and releases the secondary oocyte into the pelvic cavity, and the fimbriae sweep it into the uterine tube.
10. The collapsed follicle fills with clotted blood, which is absorbed. The granulosa and theca cells enlarge into the corpus luteum, which secretes progesterone, estrogen and inhibin.
11. Progesterone and estrogen together are strong negative feedback, helped by inhibin. LH and FSH stay low, so no new follicle matures and no second surge occurs.
12. Without a pregnancy, the corpus luteum breaks down after about 12 days. Progesterone and estrogen fall sharply, their block on FSH and LH lifts, and the next cycle begins (step 1).

- How sustained estrogen flips the feedback is not fully worked out; estrogen-sensitive hypothalamic neurons that drive GnRH are involved, and the pituitary becomes more responsive.
- The surge ends on its own: LH turns the follicle cells toward progesterone, so estradiol dips before rupture.
- The luteal phase stays near 14 days because the corpus luteum has a fixed lifespan; the follicular phase varies. A 32 day cycle ovulates around day 18.
- Combined birth control pills give steady estrogen and a progestin, so feedback stays negative and no LH surge or ovulation occurs.

### The cycle across the days (a 28 day cycle)

| Days | Ovary | Hormones and feedback | Uterus |
|---|---|---|---|
| **1 to 5** | A group of follicles starts to grow | Estrogen and progesterone at their lowest; FSH rising a little | Menstrual phase: functional layer shed |
| **6 to 11** | Dominant follicle selected and grows; others die | Estradiol rising; with inhibin, negative feedback lowers FSH | Proliferative phase: lining rebuilds |
| **12 to 13** | Dominant follicle at full size | Estradiol peaks and stays high; feedback switches to positive; LH surge begins | Late proliferative; cervical mucus thin and watery |
| **14** | Ovulation, about 32 to 36 hours after the surge begins | LH peak, smaller FSH peak; estradiol dips | End of proliferative phase |
| **15 to 21** | Corpus luteum forms and grows | Progesterone rises to a peak about a week after ovulation; estradiol has a second, smaller peak; negative feedback holds LH and FSH low | Secretory phase: glands secrete; lining ready for an embryo |
| **22 to 28** | Corpus luteum breaks down into the corpus albicans | Progesterone, estradiol and inhibin fall; FSH begins to rise | Late secretory: spiral arteries constrict; bleeding begins on the next day 1 |

- On a cycle graph, the estradiol peak comes about a day before the LH surge, ovulation follows the surge, and the progesterone peak about a week later comes from the corpus luteum. A progesterone rise in the second half of the cycle shows that ovulation happened; in a cycle without ovulation the LH surge is missing and progesterone never rises.

### The uterine cycle, step by step

1. Menstrual phase (about days 1 to 5): the functional layer and blood leave as menstrual flow. The basal layer stays.
2. Proliferative phase (about days 6 to 14): estradiol from the growing follicles makes the basal layer build a new functional layer. It thickens, its glands lengthen, and the spiral arteries grow up into it. Estrogen also adds progesterone receptors to the endometrial cells, and it thins the cervical mucus so sperm can pass.
3. Secretory phase (about days 15 to 28): progesterone from the corpus luteum acts on the estrogen-primed lining. The glands coil and secrete a glycogen-rich fluid to feed an embryo, and the spiral arteries coil more. Progesterone thickens the cervical mucus into a plug that blocks sperm and bacteria.
4. Without a pregnancy, the corpus luteum dies and progesterone falls. The spiral arteries constrict, and the functional layer is starved of oxygen and begins to die (the ischemic phase, in the last few days).
5. The arteries relax, blood flows into the dying tissue, and the functional layer breaks away. This is day 1 of the next cycle.

- The lining is shed because progesterone goes away, not because a hormone arrives. If the corpus luteum is rescued and progesterone stays up, the lining stays.
- Midcycle cervical mucus is thin and more alkaline. It guides sperm, shields them from phagocytes and vaginal acid, supplies energy, and helps start capacitation.
- Mittelschmerz is a twinge of lower abdominal pain at ovulation, probably from stretching of the ovary wall or peritoneal irritation by fluid or blood from the follicle.
- In about 1 to 2% of cycles two oocytes are ovulated; fertilized by two sperm, they become fraternal twins. Identical twins come from one zygote that splits early.

## What estrogen and progesterone do

**Competency 19:** Uterine cycle.

**In your textbook:** Silverthorn Chapter 26, the part on female reproduction, on the hormones of the menstrual cycle, and the part on procreation, on the human sexual response.

There are three main estrogens: estradiol, estrone and estriol. Estradiol, made mostly by the ovary, is the most abundant and strongest during the reproductive years. Estrone, made mostly in fat tissue, is the main estrogen after menopause, and estriol is the main one made by the placenta in pregnancy. Estrogen comes from the growing follicles under FSH and LH, and from the corpus luteum. Progesterone comes mainly from the corpus luteum under LH. Estrogen usually prepares a tissue, often by adding progesterone receptors, and progesterone then acts on the prepared tissue.

| Target | Estrogen | Progesterone |
|---|---|---|
| **Hypothalamus and pituitary** | Negative feedback at low to moderate levels; positive feedback when high for about two days | Negative feedback |
| **Ovary** | Promotes follicle growth and ovulation | A small local rise just before ovulation helps the follicle wall rupture |
| **Endometrium** | Proliferative phase: thickens the lining and adds progesterone receptors | Secretory phase: glands secrete; maintains the lining in pregnancy |
| **Cervical mucus** | Thin and watery; lets sperm pass | Thick and sticky; a plug against sperm and bacteria |
| **Uterine tubes** | Increases fimbria activity and the beating of the cilia | Not well established |
| **Myometrium** | Raises excitability; late in pregnancy adds oxytocin receptors | Quiets contractions during pregnancy |
| **Breast** | Grows the ducts | Grows the lobules and alveoli; with estrogen brings the gland to its milk-making state |
| **Bone** | Growth spurt, then closes the growth plates; slows bone resorption | Little direct effect |
| **Kidney** | Promotes Na+ reabsorption, so less water is lost | Opposes aldosterone and promotes Na+ and water loss |
| **Other** | Female pattern of fat storage; raises HDL and lowers LDL; anabolic | Raises basal body temperature after ovulation |

- At puberty, estrogen enlarges the uterine tubes, uterus, vagina and external genitalia, thickens the

vaginal lining, grows the breasts, and widens the pelvis.

- Estrogen drives the growth spurt, around ages 11 to 12 in girls, earlier than in boys, then closes the growth plates, so girls finish growing earlier.
- During pregnancy, estrogen grows the uterus, the external genitalia and the breasts.
- Basal body temperature rises by about 0.3 °C (0.5 °F) after ovulation because progesterone acts on the hypothalamus. The rise confirms ovulation after the fact.
- Estrogen's effect on blood lipids helps protect the heart and vessels.
- Estrogens and androgens, including DHEA from the adrenal cortex, likely both contribute to female sex drive, but which hormones matter most is not settled.

### The female sexual response

- Touch and thoughts trigger the same autonomic pathways as in men: parasympathetic signals widen arterioles, and the clitoris, vestibular bulbs, vaginal walls and breasts fill with blood.
- Fluid seeps through the vaginal walls and the greater vestibular glands add mucus for lubrication.
- Orgasm brings rising muscle tension, heart rate and blood pressure and rhythmic uterine and pelvic floor contractions. There is no ejaculation or refractory period.
- Orgasm is not needed for conception.

### Amenorrhea from low energy stores

1. Very heavy training or too little food lowers body fat and energy availability.
2. Leptin from fat tissue falls, and fat tissue also converts less adrenal androgen to estrogen.
3. The hypothalamus reads low leptin as too little energy for reproduction and slows GnRH pulses.
4. LH and FSH fall, follicles do not mature, and estrogen stays low.
5. Cycles stop (amenorrhea), or menarche is delayed in a young athlete.
6. Low estrogen removes its brake on bone resorption, so even young women can lose bone and develop osteoporosis. Restoring energy balance and body fat usually lets cycles return.

## Fertilization, implantation and the hormones of pregnancy

**Competency 20:** Pregnancy and placental hormones.

**In your textbook:** Silverthorn Chapter 26, the part on procreation, on fertilization, and the part on pregnancy and parturition, on implantation and the hormones of pregnancy.

### Fertilization, step by step

1. Sperm swim from the vagina through thin midcycle cervical mucus and the uterus into the tubes, helped by uterine contractions. Only a few hundred reach the tube.
2. On the way, secretions of the female tract capacitate the sperm.
3. After ovulation the fimbriae sweep the secondary oocyte, with its corona radiata, into the tube. Cilia and peristalsis move it toward the ampulla.
4. Sperm meet the oocyte, usually in the ampulla, and push between the cells of the corona radiata.
5. A sperm binds to proteins in the zona pellucida, which sets off the acrosome reaction: the acrosome releases its enzymes, which digest a path through the zona.
6. The membrane of one sperm fuses with the oocyte membrane, and the sperm nucleus enters.
7. The oocyte releases the contents of its cortical granules, which change the zona pellucida so no other sperm can bind or enter. This block prevents polyspermy, fertilization by more than one sperm.
8. Entry of the sperm makes the secondary oocyte finish meiosis II, forming the ovum and the second polar body.
9. Within hours the nuclei of the sperm and ovum join, and the diploid cell is now a zygote.

- The oocyte can be fertilized for about 12 to 24 hours after ovulation; sperm survive up to about 5 days in fertile cervical mucus. So intercourse in the 5 days before ovulation or on that day can lead to pregnancy.
- Fertilization occasionally happens in the pelvic cavity outside the tube. Unfertilized oocytes disintegrate.

### Implantation, step by step

1. The zygote divides by mitosis (cleavage) as cilia and peristalsis move it down the uterine tube. It becomes a solid ball of cells, the morula.
2. About 4 to 5 days after fertilization it enters the uterus as a blastocyst: a hollow ball with an outer layer, the

trophoblast, and an inner cell mass that will become the embryo. It lives on uterine gland secretions for a few days.

3. The blastocyst hatches from the zona pellucida.
4. About 6 to 7 days after fertilization, the trophoblast attaches to the secretory-phase endometrium and invades it. The endometrium thickens around it.
5. The trophoblast releases human chorionic gonadotropin (hCG) into the mother's blood.
6. The trophoblast forms the chorion, whose fingerlike villi grow into pools of maternal blood. The chorion and the maternal tissue together become the placenta, where nutrients, gases and wastes cross between mother and fetus. Their blood does not normally mix.

- Implantation outside the uterus, usually in a tube, is an ectopic pregnancy. The tube can rupture, an emergency. Tubal scarring from past infection raises the risk.

### Rescue of the corpus luteum, step by step

1. hCG from the trophoblast reaches the ovary in the blood.
2. hCG is similar in structure to LH and binds the LH receptors on the corpus luteum.
3. The corpus luteum does not break down at about 12 days. It keeps making progesterone and estrogen.
4. Progesterone stays high, so the endometrium is kept and there is no period, the first sign of pregnancy.
5. Around weeks 7 to 9 the placenta takes over most progesterone production, and by the end of the first trimester it is the main source (sources differ slightly on the exact week). The corpus luteum is then no longer needed.
6. If this shift failed and progesterone fell, the lining would shed and the pregnancy would be lost.

- hCG passes into the mother's urine, which is what pregnancy tests detect. Levels rise quickly in early pregnancy, peak in the first trimester, and then fall.

| Hormone | Main source | Role |
|---|---|---|
| **hCG** | Trophoblast, then placenta | Rescues the corpus luteum; peaks in the first trimester, then falls; in a male fetus it drives the Leydig cells to make testosterone |
| **Progesterone** | Corpus luteum, then placenta | Keeps the uterine muscle quiet and maintains the lining; keeps the breast from making milk; thickens the cervical mucus plug |
| **Estrogen** | Corpus luteum, then placenta (using androgens from the fetal adrenal gland) | Grows the uterus and the breast ducts; late in pregnancy adds oxytocin receptors to the uterine muscle |
| **Human placental lactogen (hPL)** | Placenta | Makes the mother's tissues less sensitive to insulin, leaving more glucose for the fetus; helps prepare the breast |
| **Relaxin** | Corpus luteum and placenta | Thought to soften the cervix and loosen the pelvic ligaments for birth; much of what is known comes from animal studies |

- Because hPL, together with other placental hormones, raises the mother's insulin resistance, some women cannot make enough extra insulin and develop gestational diabetes.

## Labor, lactation, menopause and infections

**Competency 21:** Parturition and lactation.

**In your textbook:** Silverthorn Chapter 26, the part on pregnancy and parturition, on labor and delivery and on lactation, and the part on growth and aging, on menopause.

### Getting ready for labor

- Late in pregnancy, estrogen rises relative to progesterone and adds oxytocin receptors and gap junctions to the myometrium, so the muscle contracts as a unit.
- The placenta's own corticotropin-releasing hormone (CRH) rises toward term and may help time birth.
- The cervix softens and thins, helped by prostaglandins and possibly relaxin.

- What finally starts human labor is not fully known; the fetus, placenta and stretched uterus all seem to contribute.

### Labor as positive feedback, step by step

1. The fetal head presses down on the cervix and stretches it.
2. Stretch receptors in the cervix send sensory signals up the spinal cord to the hypothalamus.
3. Oxytocin neurons in the hypothalamus fire, and oxytocin is released from their endings in the posterior pituitary into the blood.
4. Oxytocin binds receptors on the myometrium, which contracts. Oxytocin also makes the uterus release prostaglandins, which strengthen the contractions.
5. Stronger contractions push the head harder into the cervix, the stretch increases, and more oxytocin is released. Each cycle of the loop is stronger than the last.
6. Delivery of the baby, and then the placenta, removes the stretch on the cervix, and the loop stops.
7. The uterus keeps contracting after delivery, squeezing the torn vessels at the placental site to limit bleeding.

**Positive feedback**

A loop in which the response strengthens the stimulus. It drives an event to completion and needs an outside event, here the birth, to stop it.

**Stages of labor**

Dilation (the cervix opens fully), expulsion (the baby is delivered), and the placental stage (the placenta is delivered).

- Synthetic oxytocin is used to start or strengthen labor and to reduce bleeding after delivery.

### Milk production, step by step

1. During pregnancy, estrogen, progesterone, prolactin and hPL grow the ducts and alveoli of the breast, but high progesterone keeps them from making milk.
2. Delivery of the placenta removes its progesterone, and milk production begins.
3. Prolactin from the anterior pituitary is normally held down by dopamine from the hypothalamus.
4. Suckling sends sensory signals from the nipple to the hypothalamus, which releases less dopamine.
5. Prolactin rises and makes the alveolar cells produce milk, mostly for the next feeding.
6. Frequent suckling keeps prolactin high; when nursing stops, milk production stops.

### The milk ejection reflex, step by step

1. The baby suckles, and touch receptors in the nipple send sensory signals to the hypothalamus.
2. Oxytocin neurons in the hypothalamus fire and release oxytocin from the posterior pituitary.
3. Oxytocin travels in the blood to the breast.
4. The myoepithelial cells around each alveolus contract and squeeze milk into the lactiferous ducts.
5. Milk flows to the nipple. This is the let-down, and it begins within seconds to a minute of suckling.

- Let-down can become a learned response to a baby's cry. Stress can block it, so a mother can make plenty of milk yet struggle to deliver it.
- Nursing oxytocin also contracts the uterus, helping it shrink after birth and causing cramps some women feel.
- High prolactin while nursing suppresses GnRH, so ovulation is often delayed. This is not reliable birth control.

### Menopause, step by step

1. Follicles are used up by atresia and ovulation, and none are replaced.
2. In the years before menopause (perimenopause), few follicles are left. Cycles become irregular, and some skip ovulation.
3. With fewer follicles, the ovaries make less estradiol and inhibin.
4. Their negative feedback weakens, so the hypothalamus and anterior pituitary release more GnRH, FSH and LH. FSH rises most, partly because inhibin is gone.
5. The few remaining follicles cannot respond, estrogen stays low, and cycles stop.
6. Menopause is confirmed after 12 months without a period, on average around age 51 in the United States.

- Hot flashes come from changes in the hypothalamus's temperature control as estrogen falls.
- Without estrogen's brake on bone resorption, bone is lost faster, and the risk of osteoporosis rises.
- The vaginal lining thins and makes less fluid, and the uterus and breasts shrink.
- LDL tends to rise and HDL to fall, and heart disease risk rises.

- Men have no sudden equivalent. Testosterone falls slowly with age, and sperm production continues, though at a lower rate.

### Sexually transmitted infections

Sexually transmitted infections (STIs) spread through sexual contact. Condoms lower the risk, though less for herpes and HPV, which spread by skin-to-skin contact. Many STIs cause no symptoms, so people can pass them on without knowing. Chlamydia and gonorrhea often cause no symptoms, especially in women, and can spread up to the uterus and tubes (pelvic inflammatory disease). The scarring that follows can block the tubes and cause infertility or ectopic pregnancy.

| STI | Cause | Common signs | Treatment |
|---|---|---|---|
| **Chlamydia** | Bacterium | Often none; discharge, painful urination, bleeding between periods | Antibiotics; curable |
| **Gonorrhea** | Bacterium | Often none; discharge, painful urination | Antibiotics; curable, but resistance is a growing problem |
| **Syphilis** | Bacterium | A painless sore (chancre), later a rash, fever and joint pain; untreated, it can damage the brain, heart and a fetus | Antibiotics; curable, but damage already done is not reversed |
| **Trichomoniasis** | Protozoan parasite | Often none, especially in men; yellow-green vaginal discharge with a strong odor | Antibiotics; curable |
| **Human papillomavirus (HPV)** | Virus | Often none; genital warts; some types cause cancers of the cervix, penis, anus and throat | No cure for the virus, but most infections clear on their own; the vaccine prevents the types behind most HPV cancers and warts |
| **Genital herpes** | Virus | Often mild or none; blister-like sores that can come back | Antivirals control outbreaks; no cure |